Appendix — Recommendations for Replication of the OpenRiverCam Station Design

Indonesia ORC Deployment — Spring 2026

PMI · IPB · BHLK · American Red Cross

Version 2026-08-31 (draft for internal review — not yet circulated) | Audience: IPB and BHLK — technical staff | Built 2026-08-31

Contents

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Companion to: Recommendations for Replication of the OpenRiverCam Station Design, prepared for IPB and BHLK following the meeting at Sukabumi, 21 August 2026.

Purpose and audience. The main report is written for a scientific but non-specialist reader, and keeps part numbers, interface detail and procedure out of its body. This appendix holds that material, for engineers who will build, procure or operate the units. Each section states which part of the report it supports. Nothing here changes a conclusion in the report; it provides the measurements and the mechanisms behind them.

A1. Camera firmware limitations in detail

Supports report §3.3 and R9.

A1.1 Platform and the origin of the limitations

The ANNKE C1200 is a Hikvision OEM product: Hikvision **G6 platform** hardware running ANNKE-branded firmware. Configuration is over **ISAPI**, Hikvision's HTTP REST interface — digest authentication, XML payloads, `GET` to read a resource and `PUT` to write it. Nearly every camera setting is reachable this way, which is what makes version-controlled camera configuration possible at all.

The limitations below are **not hardware limitations**. They are places where the ANNKE firmware exposes less of the interface than genuine Hikvision firmware does on the same silicon, or where the behaviour sits below the firmware entirely.

A1.2 Recorded-file retrieval is not exposed (the Profile C gap)

Four capture profiles were designed and tested. Profile C was intended to be the quality ceiling: have the camera record to its own SD card at the full configured bitrate with no real-time delivery constraint, then pull the finished file over the Ethernet link faster than real time.

The sequence requires five ISAPI calls:

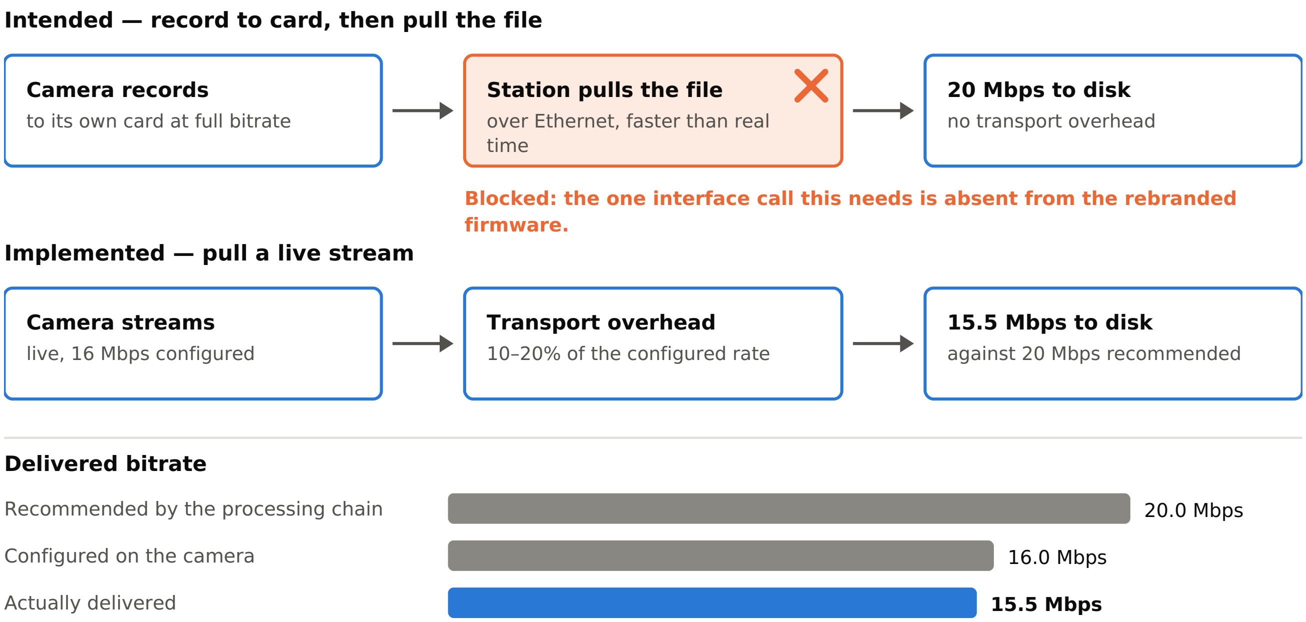
```
1. PUT  /ISAPI/ContentMgmt/record/control/manual/start/tracks/101
2.      wait N seconds
3. PUT  /ISAPI/ContentMgmt/record/control/manual/stop/tracks/101
4. POST /ISAPI/ContentMgmt/search
5. GET  /ISAPI/ContentMgmt/download?playbackURI=<uri>     <-- not exposed
```

Step 5 is the blocker. On genuine Hikvision firmware the endpoint returns the recorded file over HTTP at wire speed. **ANNKE's firmware exposes it for still images only**; recorded video files are not downloadable over HTTP ContentMgmt.

The implemented workaround pulls the clip by RTSP playback with codec copy (`ffmpeg -rtsp_transport tcp -c copy`) — a remux with no re-encode, so byte quality is preserved, but it reintroduces exactly the RTSP transport dependence Profile C existed to avoid. Profile C was therefore not selected for production.

Figure A1 The capture path: intended against implemented

The limitation is in the firmware, not in the optics or the sensor.



The intended path had the camera record to its own card and the station pull the finished file over Ethernet. The one interface call that needs is absent from the rebranded firmware, so the path is blocked and capture falls back to a live stream that pays 10–20% transport overhead.

A1.3 The measured cost in delivered bitrate

Quantity	Value
ORC / pyorc recommendation at 1080p	20 Mbps
Configured CBR target (baseline profile)	16 Mbps
Delivered over RTSP, measured	~15.5 Mbps
RTSP transport overhead	10–20% of configured bitrate
Expected gain if Profile C were available	15–25% higher effective bitrate to disk

Profiles tested were: baseline (1080p, 16 Mbps CBR, H.264, RTSP); A (1080p, 20 Mbps CBR, RTSP); B (720p, 20 Mbps CBR, RTSP — maximum bits per pixel); C (1080p, 20 Mbps VBR, local SD — blocked); and E (1080p, 12 Mbps CBR, H.265, RTSP). Profiles were compared on **PIV pass rate** — the percentage of interrogation windows passing both a cross-correlation threshold and a signal-to-noise threshold, run through the same FFPIV engine pyorc uses — rather than on generic image-quality proxies, because compression artefacts raise texture metrics without producing anything that moves with the water.

A1.4 The power-on white light (ISS-004)

The camera illuminates its white LED ring at full brightness for two to three seconds on **every** power-on. This is a hardware self-check that runs **before the operating system loads**, so no configuration reaches it.

Investigated and ruled out:

Approach	Result
`supplementLightMode=irLight` via ISAPI	Setting persists; boot flash still fires
`whiteLightBrightness=0` via ISAPI	Setting persists; boot flash still fires
`/ISAPI/System/externalDevice enabled=false`	Endpoint not supported on this firmware
Disabling smart events / motion detection	Boot flash is not event-triggered
CGI endpoints	Removed from G5/G6 firmware
Telnet / SSH	Not accessible on this firmware generation
ONVIF auxiliary control	Maps to wired relay outputs, not LED drivers
Firmware update	No Hikvision firmware version offers a boot-flash disable
Physical masking	White and IR LEDs interleave in one ring behind a single dome — masking white also blocks IR night vision
Continuous camera power	118 Wh/day → 425 Wh/day; autonomy 2.5 d → under 1 d. Not viable on the solar budget

At Sukabumi — an urban canal with residences on both banks — a 15-minute cycle would produce 96 flashes per day. The station runs a 30-minute cycle, 48 per day. The issue record lists a longer cycle among the candidate mitigations; on this camera, halving the flash rate also halves the measurement rate.

A1.5 Boot time

Milestone	Time
Basic boot, LED activity	~30 s
Full boot, network ready	~60 s
Recommended settling time	1–2 min
PoE negotiation	under 1 s

Boot draws 1–4 W above nominal, settling after roughly 30 seconds. The firmware-level options that would reduce this — suppressing console output, LZ4 rather than gzip for kernel and initramfs, asynchronous driver probing, removing unused modules, zero bootloader delay — are not exposed on consumer camera firmware. Purpose-built SoCs reach 250 ms boot, but not in this class of camera.

On a station whose normal wake is about two minutes, a 30–60 second camera boot is a large fraction of the awake period, and the energy is spent before any video exists. This is the single strongest technical argument for R11.

A2. Replacing the camera firmware

Supports report §3.4.

A2.1 What was researched, and what was not

The procedure studied was **cross-flashing genuine Hikvision firmware onto the ANNKE-branded hardware**, which share the Hikvision G6 platform. The community reports that the Hikvision image restores the full ISAPI surface including `ContentMgmt/download`, which would make Profile C viable as designed.

This is **unofficial vendor firmware, not an open-source firmware stack.** Open-source firmware projects exist for some IP camera SoCs, but none was evaluated against this hardware and nothing in this project's record supports a recommendation on them. The risks in A2.2 arise from the act of replacement rather than from the licence of what is loaded, so they apply to either route.

Status: documented as a contingency, not executed. The gate was to attempt it only if RTSP-live capture proved unable to deliver an acceptable PIV pass rate. It did not, so the procedure was never run.

A2.2 Risks

Bricking, with an expensive recovery path. Some ANNKE firmware revisions reject the Hikvision `digicap.dav` installer at the web uploader. A failed flash can leave the camera unbootable; recovery is a TFTP boot from the internal bootloader, and on some models that requires opening the housing and wiring to the UART pads. On an IP67 camera intended for permanent outdoor mounting, breaking the seal is itself a reliability event.

Hardware-revision dependence. The community recipe applies to specific ANNKE hardware revisions. The revision of each unit must be confirmed against the recipe before flashing, and a later procurement batch may not match even where an earlier one did — a poor property for a pilot buying units over time.

No support path. The authoritative procedure is a community forum thread that is edited over time, not vendor documentation. It must be re-read before every attempt and can stop being correct without notice.

Warranty void. Cross-flashing departs from the supported image; RMA is not available for unrelated later hardware faults.

Spares stop being interchangeable. A cross-flashed camera and a factory-firmware spare do not accept the same configuration profile. Every spare must be flashed to match, which hands the flashing procedure — and its bricking risk — to whoever maintains the station locally. This conflicts directly with the design constraint that any component be replaceable in five minutes with common tools.

It does not fix the boot flash. That behaviour is pre-OS; no firmware image suppresses it. Two of the three limitations in A1 survive the flash.

A2.3 If it is attempted anyway

Prerequisites, in order:

- An **uncommitted spare** as the test unit. Never flash an installed production camera first.
- Full configuration backup of the spare, pulled and committed to version control.
- The shipping ANNKE firmware image saved, for rollback.
- A Hikvision image matching the G6 platform, from the region-appropriate support site or a known-good community mirror.
- Hardware revision of the test unit confirmed against the recipe's supported revisions.
- A TFTP server prepared on a laptop for recovery, serving the recovery image at the bootloader's default address.
- Physical access to the camera body, and a USB-TTL adapter, for worst-case UART recovery.

Verification order after flashing, stopping and rolling back at any failure: web UI login; `GET /ISAPI/System/deviceInfo` returns sensible XML; an RTSP clip pulls normally (confirms encoder, sensor, PoE and network); then the make-or-break test, a recorded clip retrieved over HTTP `ContentMgmt/download`; then a profile comparison against the RTSP baseline on delivered bitrate and PIV pass rate.

A3. Survey scope of work

Supports report R2.

Required deliverables for a contracted survey at one site:

1. **Ground control points** — 6–10 points visible from the camera view, in UTM Zone 48S (SRGI2013), horizontal RMSE ≤ 3 cm, vertical RMSE ≤ 7 cm, **RTK Fixed observations only, no Float**.
2. **River cross-section** — distance–elevation pairs bank to bank through the camera view, chainage from a defined datum stake, 0.5–1.0 m spacing with extra points at every break in slope, covering wetted and dry portions.
3. **Staff gauge datum tie** — elevation of the gauge zero in the same coordinate system as the control points, with at least two independent checks.
4. **Permanent benchmark** — one concrete or rebar monument per site, outside the flood footprint, coordinates recorded, so a future team can re-establish the datum if a control point is lost.
5. **Written report** — equipment model and serial number, CORS stations used, observation duration and fix status per point, processing software, computed RMSE, and field sketches of control and benchmark positions.
6. **Data files** — point list as CSV (ID, Easting, Northing, Elevation, RMSE-H, RMSE-V, duration, fix status), raw RINEX logs where static observation was used, cross-section as CSV.

Indicative cost: Rp 5,000,000–15,000,000 per site, one site, one day of field work, all deliverables, including PPN. A quote materially below this indicates equipment that is not multi-band geodetic grade.

Acceptance checks on delivery. Every control point meets the RMSE gate — any that does not is re-measured or removed. The quote and report state RTK **Fixed**, not generic “centimetre accuracy”. One consistent vertical datum across control points, cross-section and gauge zero.

Two field conditions. Have the staff gauge installed at its final position before the survey, since the datum cannot be tied to a gauge that later moves. And be on site during the survey — reported RMSE values are only meaningful if the points measured are the ones the camera can actually see.

A3.1 Why this is specified so tightly

Two RTK surveys at Sukabumi, on consecutive days with the same equipment and methods, produced check-point spreads of approximately **99 cm horizontal and 139 cm vertical** between repeat occupations of the same physical markers, exceeding the applicable RTK gate by roughly 30 times. Same-marker drift between day one and day two reached 89 cm.

IPB’s total-station survey replaced the RTK approach and is what the station runs on. The deployed camera configuration (`Fit 6`, applied 2026-06-11) is built from IPB data alone — GCP coordinates from the IPB spreadsheet, cross-sections from the IPB transects, calibration frame from the May survey video — and fits at **0.037 m RMSE** against the 5 cm target, with `z_0 = h_ref = 615.0 m`. An interim auto-fit calibration recovered from the failed RTK data (4.61 cm RMSE on a six-point subset, `z_0 = 617.065`) was used before that and is now obsolete. **The two must not be mixed**: the IPB low-water surface is about 2 m lower. See `survey_data/ipb_survey_1/handoff_station/`.

Candidate causes of the RTK noise include poor base-station coordinate quality, sky obstruction and RF interference at the urban canal site — one of the three reasons an open site is recommended.

A4. Availability record

Supports the report’s field section.

Read this as a record of failure modes in this design, not as a performance assessment of the deployment. Sukabumi is a volunteer-supported pilot that was never built to production standards of availability or record continuity. The numbers below are here so that an engineer building a different design knows which failure modes are real and worth designing against. They are not a benchmark, and they should not be used as one.

Reconstructed from `sensor_readings` on the server: the station writes rows on every wake, so their absence bounds when it was not running. Regenerate with `liveorc_server/station-health/station_gaps.py --since 2026-04-01`.

Window: **2026-04-16 to 2026-08-28, 133.5 days observed. 13 interruptions.**

Onset (WIB)	Recovered (WIB)	Duration
04-17 06:23	04-17 11:03	4.7 h
04-17 15:01	04-17 17:10	2.1 h
04-18 11:33	04-18 12:26	0.9 h
04-19 08:20	04-19 13:17	4.9 h

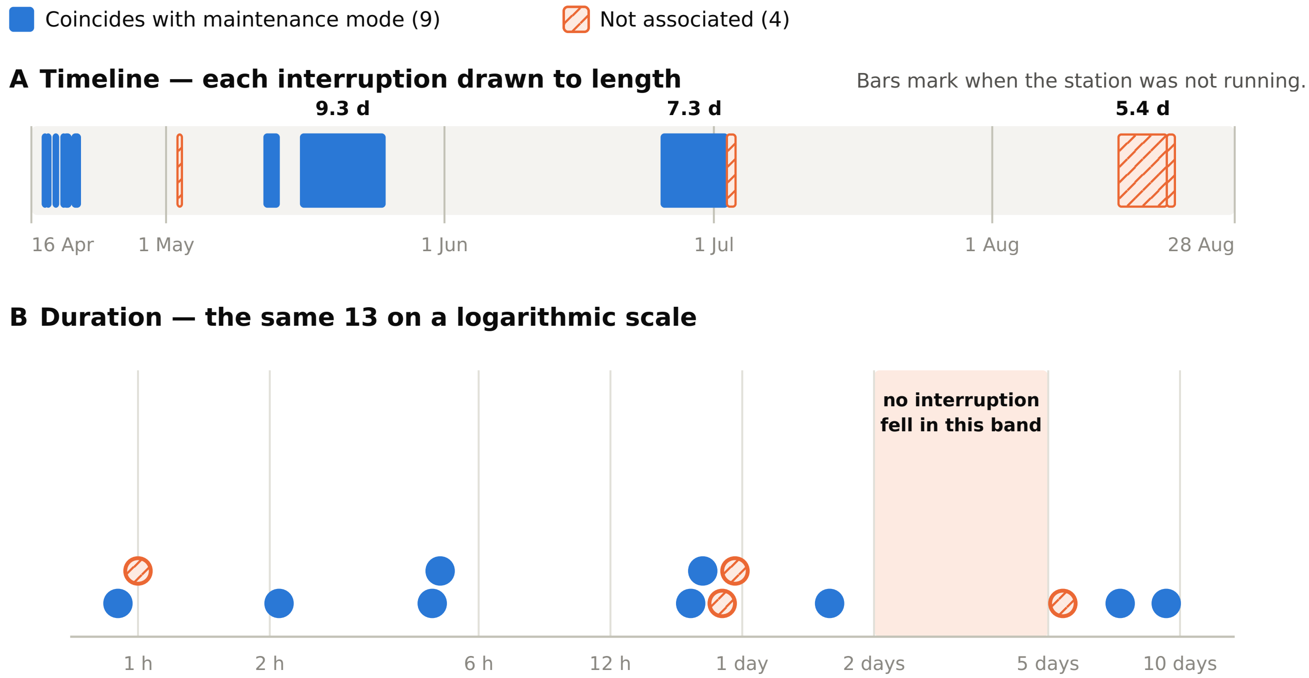
Onset (WIB)	Recovered (WIB)	Duration
04-19 14:09	04-20 08:30	18.3 h
04-20 14:13	04-21 09:43	19.5 h
05-02 06:31	05-02 07:31	1.0 h
05-11 23:01	05-13 13:03	38.0 h
05-16 00:47	05-25 07:00	**9.3 d**
06-25 04:30	07-02 10:38	**7.3 d**
07-02 10:43	07-03 09:47	23.1 h
08-15 01:30	08-20 10:39	**5.4 d**
08-20 10:39	08-21 08:14	21.6 h

Raw total 27.5 days. Roughly **2.7 days of the 05-16 interruption was a sensor-upload failure rather than downtime** — the station was running and its rows did not reach the server — so genuine downtime is approximately 24.8 days. A further interruption began 2026-08-28 and was open at the date of writing.

Duration distribution: 9 under 24 h; 1 at 24–48 h; **0 between 2 and 5 days**; 3 at 5 days and over.

Figure A2 The availability record, 16 April to 28 August 2026

13 interruptions across 133.5 days observed. The gap in the lower panel is the finding.



9 interruptions under 24 hours · 1 at 38 hours · 0 between 2 and 5 days · 3 at 5 days and over

A missed wake leaves the next-startup alarm in the past and nothing re-arms it, so an interruption either clears within a day or runs until something external restarts the station.

Panel A places each interruption across the observation window, drawn to its true length and marked by whether it coincides with maintenance mode. Panel B puts the same 13 on a logarithmic duration scale. The empty band between 2 and 5 days is the latch signature: an interruption either clears within a day or runs until something external restarts the station.

Two caveats on the method. Rows accumulate locally and backfill on reconnect, so a gap that later fills was an upload failure, not downtime; the 13 above stayed empty long after recovery. And the station's CSVs rotate at 30 days, so for gaps older than that a backlog could have been destroyed before it could backfill.

A4.1 Maintenance mode

`orc-maintenance-check` fetches a station mode file from a public repository on every boot. Set to `maintenance`, it creates a flag that makes `orc-capture` skip capture entirely and leaves the processor awake for the full scheduled ON window — roughly **12× the energy of a normal wake**, and no data.

Seven maintenance windows totalling 549 h, 17% of the observation span:

	Long-wake ticks	Rate
Inside maintenance	870	**1.59/h**
Outside	477	0.18/h

A factor of **8.8**. Nine of the 13 interruptions fall inside or immediately after a maintenance window, including the two longest.

The flag cannot be stuck, by design: it lives on tmpfs and is destroyed on every power cycle, recreated only if the remote file still reads `maintenance`, and every failure path in the check removes it. The defect is not stickiness but the absence of an expiry and of any alarm while it is set.

One inference withdrawn. An earlier analysis reported long wakes preceding outage onsets at p = 2.7e-07 and read it causally. The association is real; maintenance mode was generating both terms. The chain from extended runtime to drain to interruption survives on the natural experiment above, not on that correlation.

A5. Optical water-level detection

Supports report §4.3 and R1.

Window 2026-07-08 to 07-14, two sets of 100 videos each (most recent by status). Station clock is UTC, verified against the tracked capture configuration; local WIB = UTC+7.

Every failure ends identically — a waterline is found, then rejected by the signal-to-noise gate:

```
Found water level at h: 614.795 m with too low signal-to-noise: 1.306 < 2.000
Water level could not be estimated from video.
```

Set	n	S/N min	median	max
Passed	100	2.004	4.009	5.561
Failed	100	1.185	1.630	1.983

The gate defines the split, so the absence of overlap is definitional and carries no information. What the distributions do show is how far each population sits from the threshold. **Failures are not marginal:** median 1.63, and only 23 of the 100 reach 1.8, so dropping the gate to 1.8 would recover under a quarter of them while admitting water levels the detector had no confidence in. **Do not lower it.**

The converse deserves equal weight and cuts the other way: **36 of the 100 passing captures fall between 2.0 and 3.0**, and 26 fall below 2.5. A substantial share of accepted water levels sit close to the threshold, which is an argument for the independent reference in R1 rather than for adjusting the gate. Figure 2 of the main report plots both distributions.

The failures find the right waterline. Detected `h` is identical across both sets (median 614.794 m against a 614.3–618.5 m search band), and only 15 of 100 failures pin near the band floor. So the search band is correct and the river was stable all week; the detection simply lacks confidence.

Hour-of-day distribution (WIB):

```
00–05   fail 0   pass 36      12      fail  8   pass 1
06      fail 3   pass  3      13      fail  4   pass 3   <- solar noon
07      fail 4   pass  4      14      fail  3   pass 4
08      fail 9   pass  0      15      fail 10   pass 2
09      fail 11  pass  0      16      fail 12   pass 1
10      fail 11  pass  1      17      fail 11   pass 2
11      fail 9   pass  3      18      fail  5   pass 4
                              19–23   fail  0   pass 44
```

Zero failures at night. All failures inside the daylight window, with **two peaks (08–12 and 15–17) and a dip at solar noon**.

Interpretation. The daylight bound implicates direct sunlight; the twin peaks with a midday dip are the geometric signature of **specular glint** — a mirror reflection off the water surface into the camera — because veiling glare tracks sky brightness rather than sun angle and would not produce the midday dip. Alternatives considered and argued against: threshold too strict (bimodality); wrong search band (identical `h` in both sets); general daytime dimness (would be monotonic from dawn, not twin-peaked); afternoon wind ripple (explains 15–17, not 08–12); weather or turbidity (would not respect the daylight boundary).

This remains a hypothesis. Direct visual confirmation — measuring the saturated-pixel fraction in the water band on matched failing and passing frames — is pending. R1 does not depend on it: the failure is measured either way, and water-level estimation aborts the whole processing run, so each daytime failure costs the entire discharge measurement.

Sampling caveat. Each set is the 100 most recent of its status, so the passing set reaches back only to 07-11. A naïve per-day ratio for 07-08 to 07-10 is a sampling artefact. The day/night and S/N results are unaffected.

A6. Data delivery

Supports report §4.4 and R7.

Measured on the station 2026-08-27, ORC-OS video table, 2026-04-08 to 2026-08-27:

	Count	Share
Total rows	5,406	
Processing: DONE	2,957	
Processing: ERROR	2,324	**43% failed**
Processing: NEW	125	
Sync: SYNCED	2,536	
Sync: FAILED	2,744	**51% never reached the server**
Sync: LOCAL	126	

Disk state at the same time: 51 GB used of 58 GB, **5.1 GB free against a 5.0 GB purge threshold** — pinned to the line rather than merely full. The disk manager checks every 300 s, purges just enough to clear the minimum, and is back under it by the next check. Files on disk reached back only to 06-07 while table rows reached to 04-08; roughly 2,190 rows had no file behind them.

A separate **12-day gap in the video record (2026-07-30 to 08-11)** occurred while sensor data flowed normally at 48 rows/day throughout. It is very likely server-side.

The lost video itself does not matter — Sukabumi is a technology pilot contributing to no operational product, so this is not a data-rescue exercise. The finding is the 51% sync failure rate as a verdict on the design, which would be disqualifying in a deployment anyone relied on.

A7. Power, scheduling and the always-on comparison

Supports report R6, R10 and R11.

A7.1 Corrected power budget

	Original estimate	Corrected
Daily consumption	~94 Wh/day	**~118 Wh/day**
Battery autonomy, no sun	3.2 days	**2.5 days**

The 25% shortfall was the quiescent draw of two DC/DC converters (~0.5 W each, continuous) and the rain gauge (~150 µA, continuous) — always-on loads wired directly to the battery bus that draw whether the Pi is awake or not. Solar margin remains adequate for normal operation; the consequence is that extended cloud exhausts the battery about

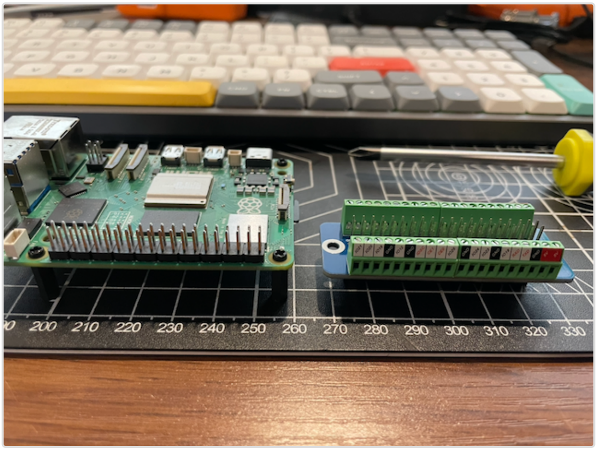
17 hours sooner than planned, which matters for setting alert thresholds and maintenance windows.

A7.2 Scheduling: Witty Pi against the Pi 5 native RTC

	Witty Pi 5 HAT+	Pi 5 native RTC
Cost	~USD 46–50 + CR2032	Included; ML-2020 cell ~USD 5
Boards in stack	One extra	None
GPIO consumed	None (I^2C only)	None
Power to Pi when off	**Cut entirely**	Standby draw remains
Input range	**6–30 V direct from battery bus**	Requires regulated 5 V
Low-voltage cut-off	**Yes, configurable**	No
Temperature cut-off	**Yes**	No
Dual input with failover	**Yes**	No
Re-arming the next wake	Separate scheduler; **can be left stale**	Written by the OS at shutdown

The native RTC was the original design for both stations. The Witty Pi was reinstated late in the build for one reason: **the Pi 5's ML-2020 RTC battery connector, a small JST-SH part, failed on both boards.** Treat that connector as handling-sensitive, provide strain relief on the battery lead, and hold a scheduling board in spares.

The right split is by power source. On **mains power or indoor compute**, the native RTC is preferable — the Witty Pi's protections are not needed and the re-arm becomes part of the normal shutdown path. On a **solar, battery-backed field station**, keep the Witty Pi: its low-voltage and temperature cut-offs and its wide-input conversion are doing real work against exactly the deep-discharge failure mode this deployment has been chasing.



The compute board and the GPIO screw-terminal riser it stacks onto. Every signal leaves as a screw terminal, which is what lets a non-specialist rewire the station with a screwdriver — and what makes the scheduling board a swap rather than a rebuild. This is the assembly R8 would move indoors and R10 would simplify; Figure 3 of the main report compares the two arrangements.

A7.3 Duty-cycled against always-on

	Solar, duty-cycled	**Mains, always-on**
Materials cost	USD 1,340	**USD 1,030**
Achievable time step	30 min	Down to continuous
Camera boot paid per measurement	30–60 s	**None after installation**
White-light flash	48 per day	**Once, at installation**
Camera power	Cycled; 118 Wh/day system total	425 Wh/day equivalent, irrelevant on mains
Missed-wake latch	**Present**	No schedule to miss
Remote diagnostic window	Tens of seconds per cycle	Continuous
Siting constraint	Anywhere with sun	**Within reach of reliable mains**

The always-on configuration is cheaper in materials and removes five distinct failure modes. Its cost is siting: it must be within reach of reliable mains power, which constrains where the station can go and may conflict with the preferred measurement section. Where mains is present but unreliable, add a UPS sized to the observed outage duration and treat the station as always-on with brief interruptions rather than as duty-cycled.

A8. Source documents

Topic	Path
Bill of materials, Sukabumi	`spring_2026_ID/BOM_Sukabumi.md`
Bill of materials, verified reference	`rc-box/BOM_VERIFIED.md`
Design specification	`rc-box/DESIGN_SPECS.md`
Split camera/compute architecture	`spring_2026_ID/docs/SPLIT_ARCHITECTURE_DESIGN.md`
Camera ISAPI configuration management	`spring_2026_ID/research/camera_isapi_config_management.md`
Cross-flash contingency research	`spring_2026_ID/research/annke_hikvision_crossflash_research.md`
Camera profile test procedure	`spring_2026_ID/docs/CAMERA_PROFILE_TEST.md`
Camera boot-time research	`rc-box/research/ip_camera_boot_times.md`
12MP IP67 camera survey	`rc-box/research/ip_cameras_12mp_ip67.md`
Witty Pi 5 research	`spring_2026_ID/research/witty_pi_5_research.md`
Survey scope of work	`survey/outsourced_survey_brief.md`
Survey error analysis	`survey/research/professional_surveyor_and_escape_hatch.md`
Optical water-level finding and dataset	`spring_2026_ID/findings/optical_wl_daytime_glint.md`
Field issue log	`spring_2026_ID/ISSUE_LOG.md`
Availability analysis tool	`spring_2026_ID/liveorc_server/station-health/station_gaps.py`
Indonesian hydrometric standards	`spring_2026_ID/research/indonesia_hydrometric_standards.md`
Lessons learned	`spring_2026_ID/LESSONS_LEARNED.md`